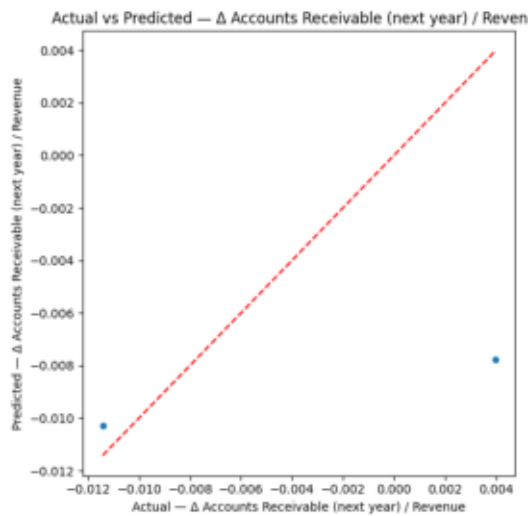


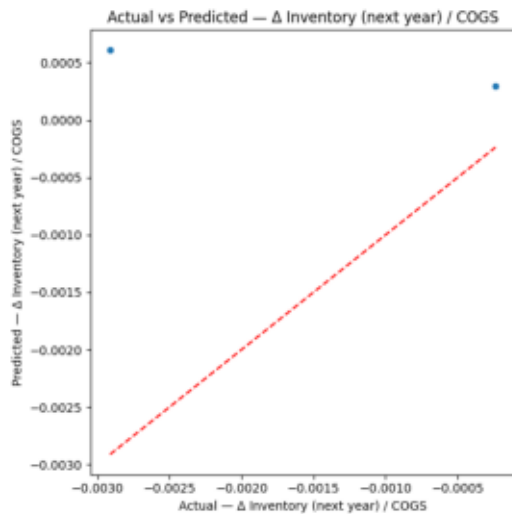


Δ Accounts Receivable (next year) / Revenue

For target Δ Accounts Receivable (next year) / Revenue, the top drivers are: Current Ratio (mean $|\text{SHAP}|=0.00258$), Accounts Payable Days (AP Days) (mean $|\text{SHAP}|=0.00122$), Operating Margin (mean $|\text{SHAP}|=0.00102$).

MAE: 0.006420 RMSE: 0.008327 R2: -0.17

2018: Actual \$265 Pred \$-518 Δ \$-783
2019: Actual \$-773 Pred \$-697 Δ \$76

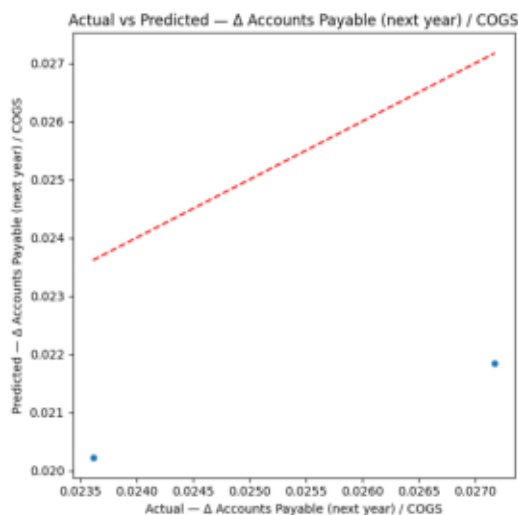


Δ Inventory (next year) / COGS

For target Δ Inventory (next year) / COGS, the top drivers are: Inventory (mean $|\text{SHAP}|=0.00016$), Accounts Payable (mean $|\text{SHAP}|=0.00016$), Δ AP (t) / COGS (mean $|\text{SHAP}|=0.00012$).

MAE: 0.002026 RMSE: 0.002518 R2: -2.54

2018: Actual \$-8 Pred \$10 Δ \$18
2019: Actual \$-100 Pred \$21 Δ \$121



Δ Accounts Payable (next year) / COGS

For target Δ Accounts Payable (next year) / COGS, the top drivers are: Revenue (mean $|\text{SHAP}|=0.00214$), Accounts Receivable (mean $|\text{SHAP}|=0.00135$), Days Sales Outstanding (DSO) (mean $|\text{SHAP}|=0.00133$).

MAE: 0.004362 RMSE: 0.004469 R2: -5.31

2018: Actual \$916 Pred \$736 Δ \$-180
2019: Actual \$811 Pred \$695 Δ \$-116

Ratio Definitions

Ratio Definitions & Human-Readable Names □ This file lists the ratio variables used in the project, their written names, short definitions/formulas, and units. - ``ΔAR_t+1_ratio`` — **`**Δ Accounts Receivable (t+1) / Revenue**`** - Definition: $(\text{Accounts Receivable}_{t+1} - \text{Accounts Receivable}_t) / \text{Revenue}_t$ - Units: dollars change normalized by Revenue (fraction) - Note: Positive → AR increases relative to revenue (collection deterioration) - ``ΔInventory_t+1_ratio`` — **`**Δ Inventory (t+1) / COGS**`** - Definition: $(\text{Inventory}_{t+1} - \text{Inventory}_t) / \text{COGS}_t$ - Units: dollars change normalized by COGS (fraction) - Note: Positive → Inventory buildup relative to cost of goods sold (cash tied up) - ``ΔAP_t+1_ratio`` — **`**Δ Accounts Payable (t+1) / COGS**`** - Definition: $(\text{Accounts Payable}_{t+1} - \text{Accounts Payable}_t) / \text{COGS}_t$ - Units: dollars change normalized by COGS (fraction) - Note: Positive → Payables increase relative to COGS (supplier financing) - ``DSO`` — **`**Days Sales Outstanding (DSO)**`** - Definition: $\text{Accounts Receivable} / \text{Revenue} * 365$ - Units: days - Note: Higher DSO → slower collections - ``Inventory_Turnover`` — **`**Inventory Turnover (times)**`** - Definition: $\text{COGS} / \text{Inventory}$ - Units: times per period - Note: Lower turnover → slower inventory movement - ``AP_Days`` — **`**Accounts Payable Days (AP Days)**`** - Definition: $\text{Accounts Payable} / \text{COGS} * 365$ - Units: days - Note: Higher AP days → longer payment terms (short-term cash benefit) - ``Operating_Margin`` — **`**Operating Margin (EBIT / Revenue)**`** - Definition: $\text{Operating Income} / \text{Revenue}$ - Units: fraction (ratio) - Note: Higher margin → more operating cash buffer - ``Current_Ratio`` — **`**Current Ratio (Current Assets / Current Liabilities)**`** - Definition: $\text{Current Assets} / \text{Current Liabilities}$ - Units: ratio - Note: Higher ratio → greater short-term liquidity - Lagged Δs (used as lag features) - ``ΔAR_t`` — **`**Δ Accounts Receivable (t) / Revenue**`** (prior period) - ``ΔInventory_t`` — **`**Δ Inventory (t) / COGS**`** (prior period) - ``ΔAP_t`` — **`**Δ Accounts Payable (t) / COGS**`** (prior period) --- File created at ``docs/RATIO_DEFINITIONS.md``. Let me know if you want these exported into ``outputs/reports/`` as a CSV or included in the PDF one-page summary.